

selfupdate

Branch Status Synthesis

The Pierre Menard Program: writing a specific text into the weights of a language model by self-distillation of privileged context.

Stage 1: memorize La tierra de Alvargonzález (Machado, 1912) with Qwen3-0.6B and up. Stage 2: Don Quijote at 120B-class scale.

github.com/arivero/selfupdate

Compiled 2026-07-06 from runs/, EXPERIMENTS.md and generated artifacts of both working trees

Reading rule: arms whose readout trains logits toward the reference text (task-CE / tail-CE) are LABELED SUPERVISED BASELINES throughout — classical distillation-plus-labels, not method results.

1. Repository & branch topology

The repository hosts one research program split into two doctrinally separated branches. Both share the same skeleton (masking, teacher caches, evaluation, scheduler tooling) but are forbidden to mix methods.

BRANCH layerwise (working tree selfupdate_lw, default branch on GitHub)

Layerwise forward distillation: each transformer block learns to reproduce the teacher's hidden state at aligned token positions, with bounded-depth credit (sliding k-connected windows).

Whole-network logit distillation is out of scope. Two experimental campaigns are CLOSED (2026-07-03/04 and 07-04/05), with ~120 runs recorded in runs/results.md and a queued Campaign 3. The training-target law (owner, 2026-07-05) governs the branch: every behavioral training term must be TEACHER-SOURCED; training logits toward the original text is task supervision and exists only as a labeled baseline.

BRANCH classic_kd (working tree selfupdate_kd)

Classical KL-based self-distillation, done properly and only that: training is teacher-logit KL, nothing that reads corpus tokens as labels. The standing question is localization: which layers are modified, which layers make the text readable, and how that moves with scale. 4 principal runs recorded, three at 14B-30B scale, one still training at snapshot time — including the program's strongest pure result so far (Qwen3-30B-A3B, CER 0.000).

The experimental setup shared by both: the same model plays teacher and student. The teacher prompt contains privileged context (a RAG passage or a visible think trace); the student receives the same prompt with the context hidden. Distillation transfers what attention extracted from the privileged block. Success is free-run recitation of the full corpus (CER: character error rate; line-exact fraction), balanced against forgetting (general-text cross-entropy drift) and, on the layerwise side, a category-resolved destruction battery (5 genres x 8 texts, benchmarks, intrusion-bait prompts).

Method contrast in one line each:

- layerwise: hidden-state matching per block (vocab_mse Gram-metric champion) with bounded, depth-uniform credit; behavioral signal, where used, is teacher-sourced (teacher_kl); the vocabulary (embeddings + LM head) is NEVER trained.
- classic_kd: full-model (or LoRA) teacher-logit KL; the question is not only whether it works but WHERE it writes.

Doctrinal walls (owner directives, enforced in configs and validators):

- On layerwise, every behavioral training term must be teacher-sourced. Task-label CE (training toward the reference text) is the supervised conflation — the training target and the eval reference become the same object — and is admitted only as a labeled baseline/ablation. Tail-only windows are banned outright ("the tail wearing a costume").
- On classic_kd, any objective reading corpus tokens as labels is out of scope by definition.
- Frozen-Vocabulary Principle on layerwise: embed/norm deltas verified bitwise 0.0 on trained checkpoints.

Owner's framing note for this synthesis (2026-07-06): the CE/tail arms scattered through the layerwise campaigns are bad implementations of traditional KL camouflaged as innovations. They appear below only where needed as gray reference points; no conclusion in this report rests on them.

Snapshot state (2026-07-06):

- layerwise: Campaign 2 closed. Storage physics well established (redundant upper-middle band, loss-dependent solutions, frozen vocabulary verified). Central pure finding: the last-3% law — distribution matching converges to teacher fidelity exactly; verbatim recall lives in the residual (SEVEN independent replications). Campaign 3 queue defined; pinned confound-resolution twins draining unattended.
- classic_kd: Qwen3-30B-A3B reached perfect probe recitation under pure teacher-logit KL (CER 0.000, line-exact 1.0); Qwen3-14B best CER 0.016 after 60 epochs; Qwen3.6-27B early (epoch 6). Base 0.6B control: CER 0.932.

2. layerwise — what the campaigns established (pure results)

2026-07-03 -> 07-05 · ~120 arms at 0.6B-14B · EXPERIMENTS.md campaigns 1-2

GOAL. Find a layerwise loss that trains with bounded backward depth and still produces behavior: recites under full-corpus eval, preserves block locality except explicitly bounded windows, has measurable forgetting cost, and scales to online-teacher LoRA and one-block-at-a-time training.

STORAGE PHYSICS (the branch's solid ground; causal probes, loss-agnostic):

- Storage is distributed and REDUNDANT across the upper-middle stack. Weight-delta mass concentrates at L22-24 (of 28, Qwen3-0.6B) in every arm, yet single-layer ablation there barely hurts (ablate L23/L24 -> CER 0.00). Peak deposits sit at ~80% depth and stay fractionally constant across scales.
- The storage loss decides WHAT is written, not just how fast. vocab_mse (Gram metric $\|W \cdot \Delta_h\|^2$, measuring hidden deltas through the frozen vocabulary geometry) writes a substantially different weight-space solution (delta-vector cosine 0.48-0.70 against all other losses) — and the best one: best recall, least forgetting, and a PORTABLE storage format (chimera checkpoints mixing bodies and tails train-free). nmse/huber ride one shared trajectory (cosine 0.95+); cosine and huber dominated.
- The Frozen-Vocabulary Principle holds bitwise: embed/norm deltas exactly 0.0 on trained checkpoints. The vocabulary is a per-layer measurement device, never a training target.
- Strict hidden matching alone stores signal but does not recite (CER ~0.85, line-exact 0). Storing is not speaking — see the last-3% law below for the pure statement of the gap.
- Anchor-KL regularization (teacher-sourced) cuts collateral damage at zero recall cost, but protection follows the anchor set (Anchor-Goodhart law): diversify anchors or be fooled. Multi-genre anchors improved every damage category simultaneously.
- Maieutic v4 data (catechism-style elicitation frames) cures elicitation brittleness (dialogue CER 0.921 -> 0.000): trigger diversity is a data property, not a loss property.

THE LAST-3% LAW (C2-34 — the branch's central pure finding, SEVEN independent replications):

Pure distribution-matching — hidden matching plus teacher-KL readout, no reference-text labels anywhere — converges to TEACHER FIDELITY exactly. The student becomes as good as the teacher's own with-context distribution, and verbatim recall lives in the residual the teacher itself lacks. At 0.6B this leaves free-run CER near 0.80 while teacher-forced fidelity is high (96.5% in one replication): the knowledge is stored and the distributions match, but greedy free-running does not lock onto the reference text at small scale. Replications: slide8kl, k6-KL, think-KL, ch8-KL (prose — mildly softer collapse, verbatim tolerance differs by genre), thinkselkl, seed-43, and the confounded-PP2 run. The law is a precise characterization, not a failure report — and the classic_kd branch shows the residual CLOSES at scale (30B-A3B: CER 0.000 under pure KL).

DISTRIBUTION-SHAPED LOSS SAFETY (C2-17/23/31): fisher and lens_kl amplify intrusion grooves (57.5% / 90% intrusion); vocab_mse and nmse are safe. Loss choice matters for cleanliness, not only recall.

SUPERVISED BASELINES (gray throughout this report). The campaigns also ran arms whose readout trains logits toward the reference text (tail-CE / task-CE, including Campaign 1's "final recipe" and Campaign 2's slide-CE slate). Per the owner's classification these are traditional KL-plus-labels in camouflage; they recite well (CER 0.007-0.015) precisely because they are supervised on the answer, and several fail the destruction battery (the C1 recipe: 16% intrusion, -5.5 benchmark). They serve as reference points for what supervision buys, nothing more.

3. layerwise — laws, hygiene, and Campaign 3

EXPERIMENTS.md C2-1..C2-35 · CLAUDE.md Current Pointer

THE LAWS OF CAMPAIGN 2 (abridged; supervised-baseline-derived items omitted):

1. Anchor-Goodhart: regularizers protect where anchored; diversify or be fooled.
2. Dilution: co-residence is free; more content and more params-per-item LOWER intrusion; batching beats drilling (the overtraining law).
3. Mimicry: matching the teacher's with-context trajectory near the readout installs the intrusion groove; mimicry-free credit is clean.
6. Reader/writer: the teacher is distilled as a READER; students exceed its writing 30-70x; teacher ceilings flat in scale (0.45-0.67 CER with the passage in context, at every size 0.6B-14B). Consolidation beats prompting.
7. Loss safety: distribution-shaped hidden losses amplify grooves; vocab_mse/nmse safe.
8. Premise condition: teacher_kl transmits only what the teacher knows in context; premise-check per data mode (think-KL collapse confirmed the prediction).
9. The last 3%: pure distribution matching converges to teacher fidelity exactly (see previous page).
10. Parallelism: PP2 evals validated (+6% cost at 0.6B); TP2 mechanical but +72% at 0.6B (collectives dominate; fat matmuls at 27B are where TP amortizes); full-FT 4B via offload_adam works at 2.3 s/item.

THE CONFOUND THAT STRUCK TWICE (experimental hygiene, 2026-07-05). A config DEFAULT (the readout-kind knob) flipped mid-campaign silently forked every queued arm that had not pinned it.

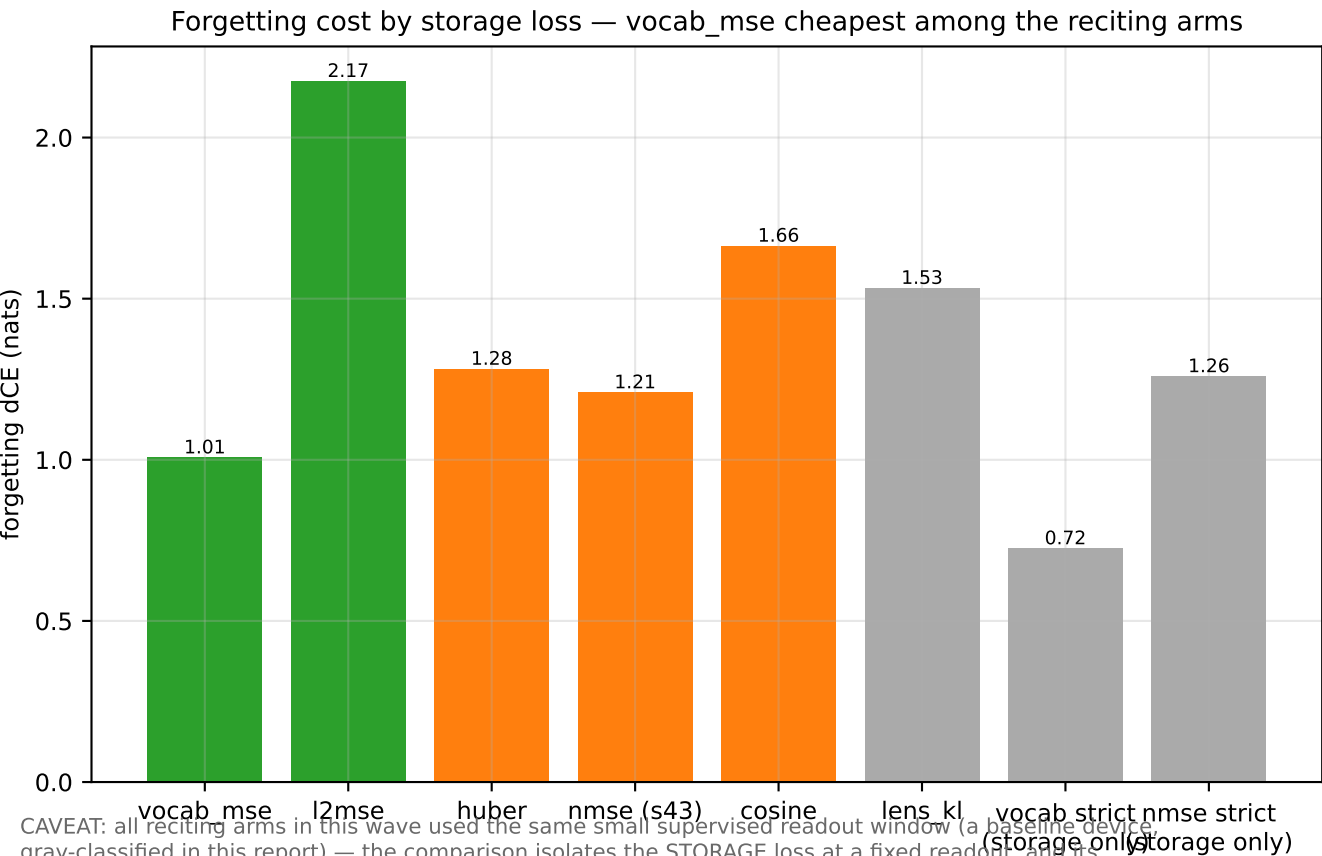
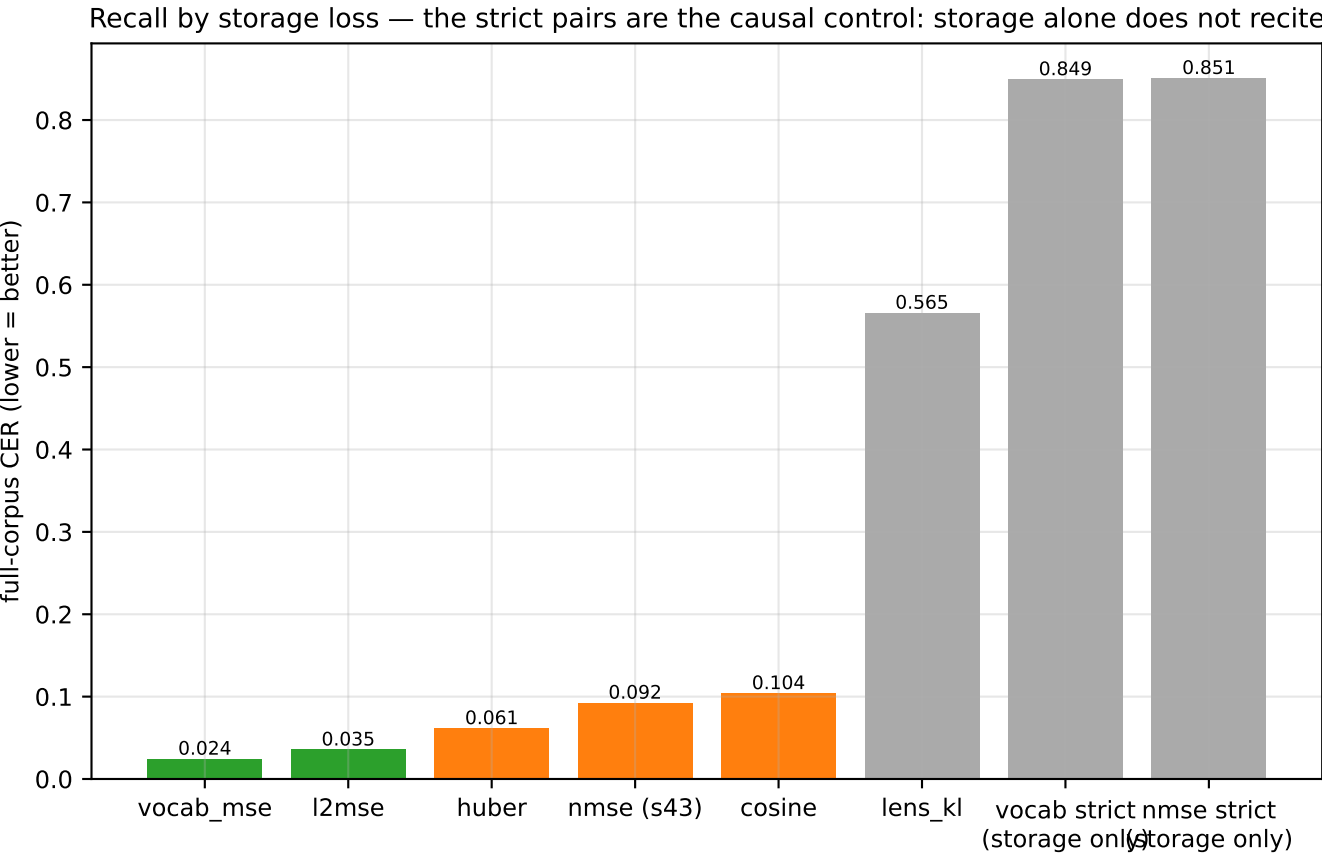
Consequences taken honestly: one finding retracted pending a pinned twin; several arms relabeled as KL-family points — which serendipitously multiplied the last-3% law's replications from two to seven. Tooling fix: the default is now an UNSET sentinel; the validator refuses any windowed run without an explicit choice; base.yaml carries the doctrinal default explicitly; 89/89 tests pass. "Defaults are experiment variables" is now ENFORCED, not remembered. The config dump is the lab notebook; the sentinel makes it sufficient.

WEIGHT-DELTA ANATOMY OF THE 0.8-CER BAND. Four arms with the same headline CER ~0.8 have four different anatomies. The pure-KL arm's weight profile is IDENTICAL to the best-reciting arm's (peak L20, 0.0311 vs 0.0310; same body/window means): pure KL wrote the same storage, only readout sharpening is missing — the last-3% law read directly off the weights. By contrast lenonly is early-layer vandalism (peak L6) and the teacher-stream arm writes deep grooves without readout. Same number, different physics; only the probes distinguish them.

SCALE & CORPUS LADDER STATE. Prose ladder (Quijote chapters, 0.6B): recall never saturated with corpus size; the destruction envelope, not capacity, is the constraint. Dilution confirmed: intrusion FALLS as the memorized corpus grows (715-verse poem 22.5% -> 8-chapter prose 5.0% on matched bait). Scale: LoRA vehicle closed at 4B (benchmark-destructive at r16 and r64); full-FT 4B via offload_adam solves the MEMORY problem; the pure recipe at scale is exactly the open Campaign 3 question, with the classic_kd 30B result as the existence proof from the sibling branch.

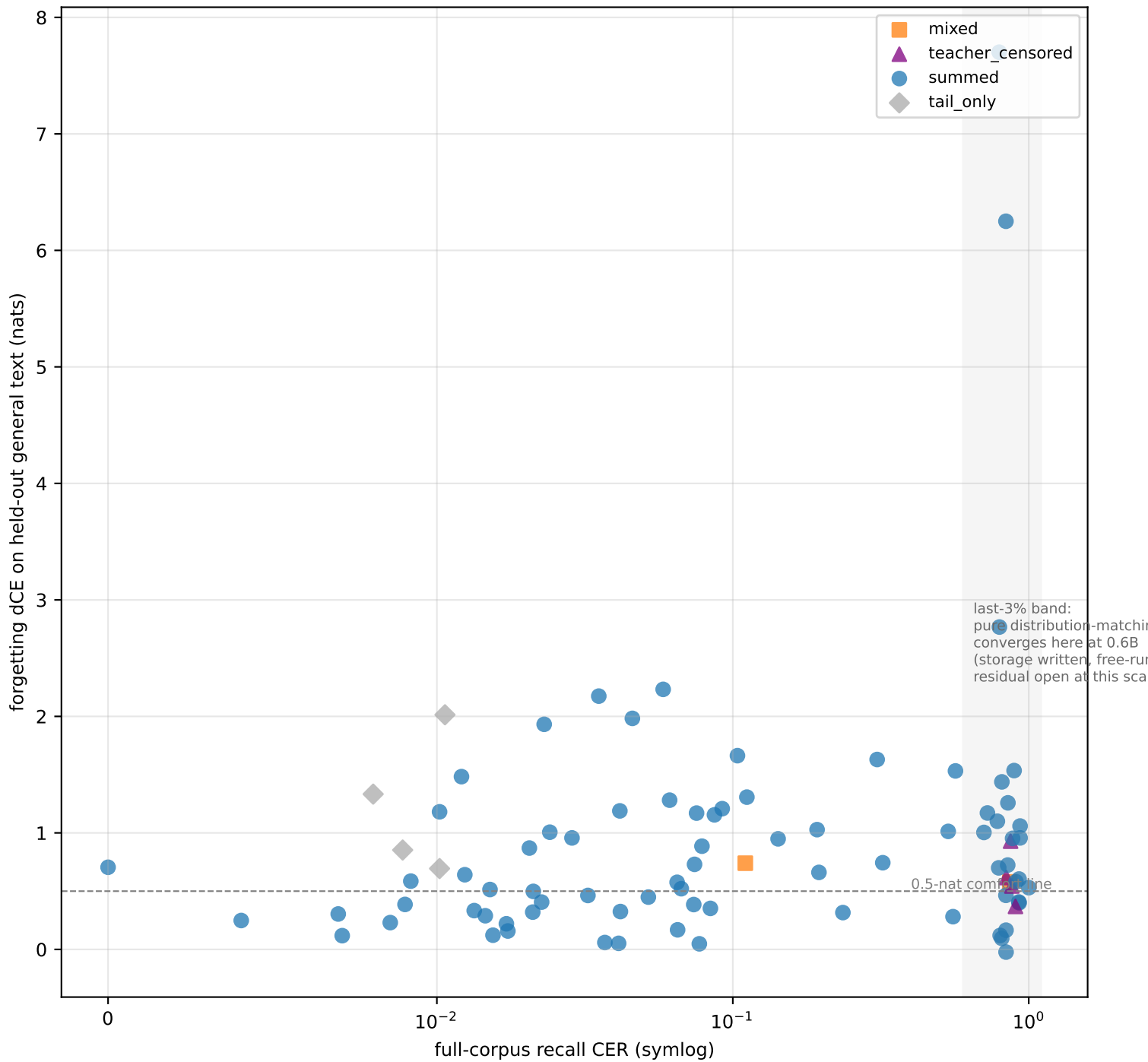
CAMPAIGN 3 QUEUE (defined, scheduler draining inheritance): teacher-stream k-windows; premise-gated thinking teacher_kl; Qwen3.6-27B bridge grid + Gemma4-E4B (embed-scaling adapter); wide-channel ragchannel; hot-loop batching+sync fix (4-6x expected — the trainer is measurably sync-bound); reincarnation; checkpoint-time layer-residuals eval mode. Pinned twins resolving the two confounds land unattended in runs/*/eval/.

4. layerwise — Wave I storage-loss comparison (0.6B, matched 13.3k items)



CAVEAT: all reciting arms in this wave used the same small supervised readout window (a baseline device, gray-classified in this report) — the comparison isolates the STORAGE loss at a fixed readout, and its verdicts (vocab_mse champion; nmse/huber one trajectory; lens_kl killed at 5x compute) were later confirmed independent of readout by the weight-delta anatomy: the pure-KL arm writes the same storage as the best reciting arm. The strict bars are the pure-storage control: signal present in the weights, no recitation.

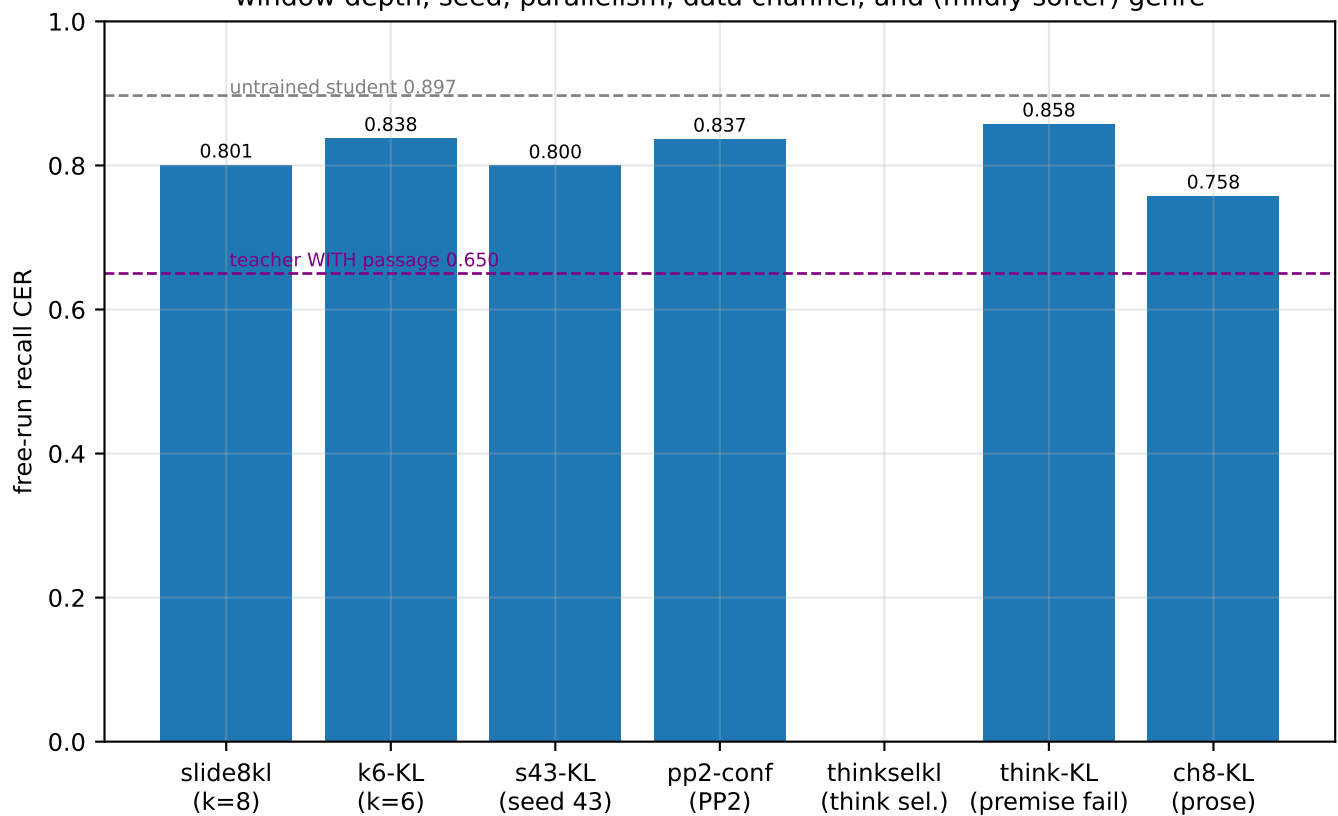
5. layerwise — the whole campaign at a glance: recall vs forgetting



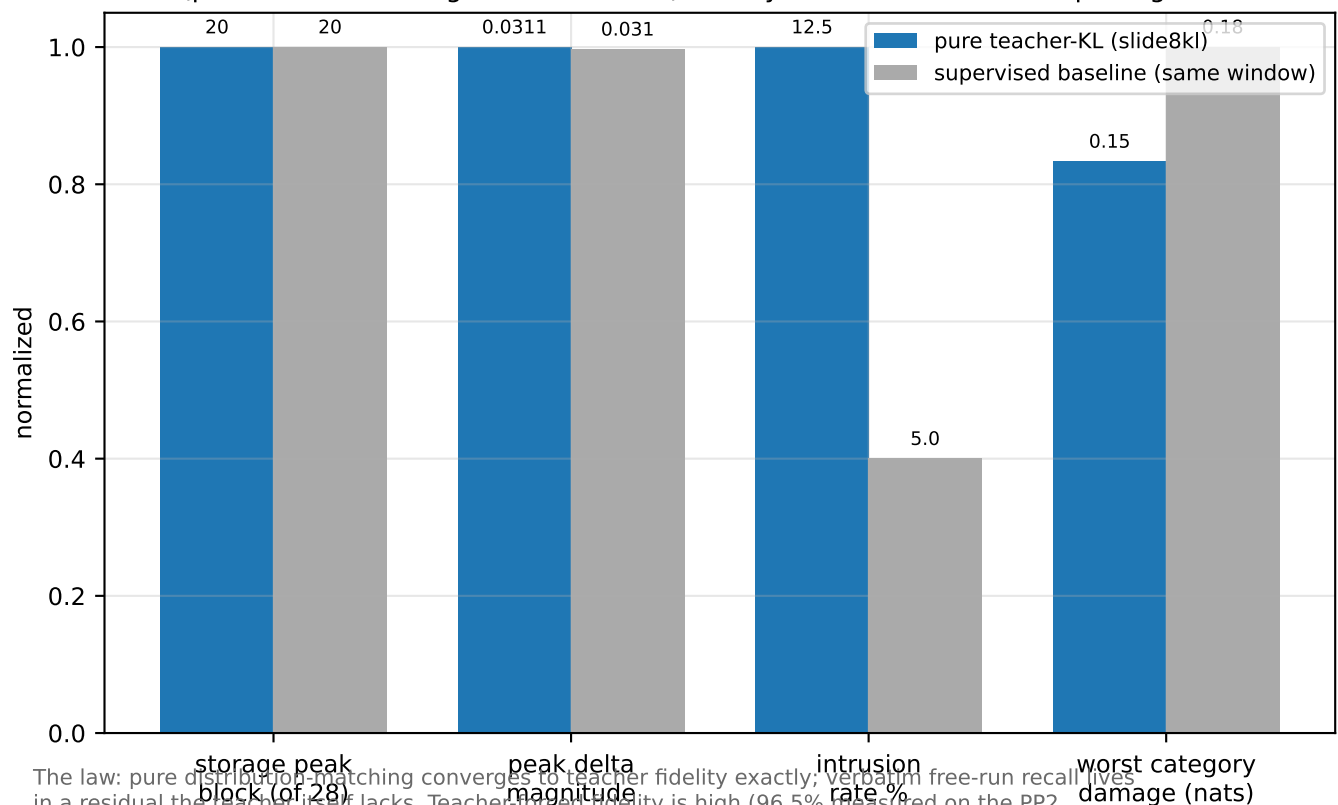
Every run in runs/results.md with both a full-corpus CER and a forgetting measurement (119 runs). The bottom-left corner is inhabited almost exclusively by supervised-readout baselines (they recite because they train on the answer); read them as reference points, not results. The scientifically loaded region is the shaded band: pure distribution-matching arms converge there at 0.6B with LOW forgetting and full storage written — the last-3% law. The tail_only markers (gray diamonds) are the banned classical-distillation shape, kept as labeled ablations only.

6. layerwise — the last-3% law: seven replications of one number

Pure teacher-KL arms at 0.6B: convergence to the teacher-fidelity plateau, independent of window depth, seed, parallelism, data channel, and (mildly softer) genre

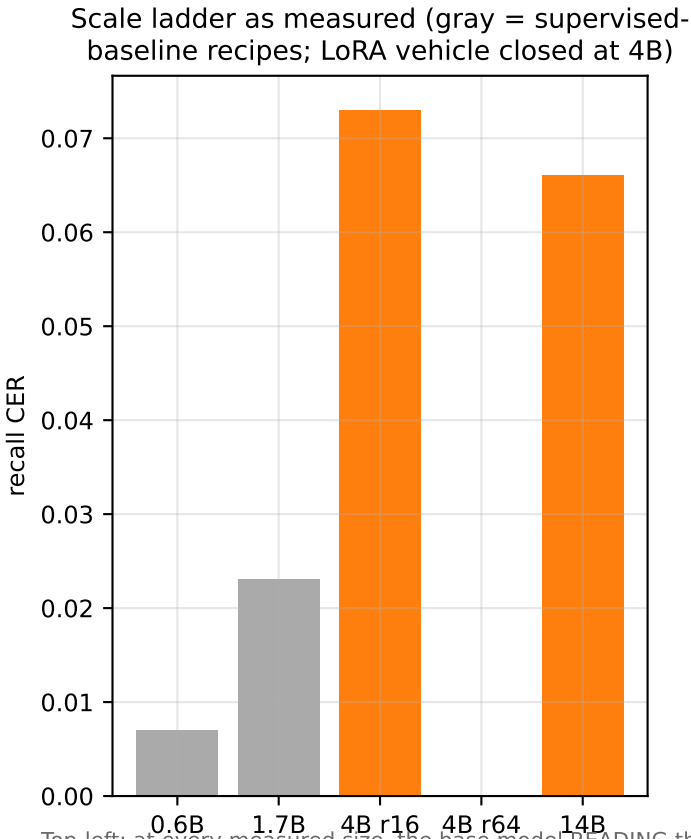
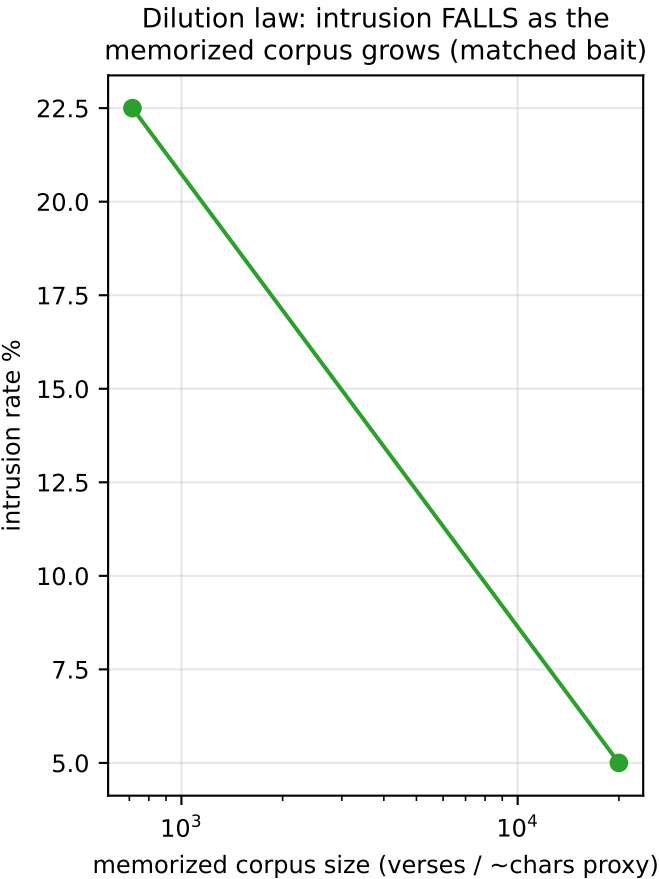
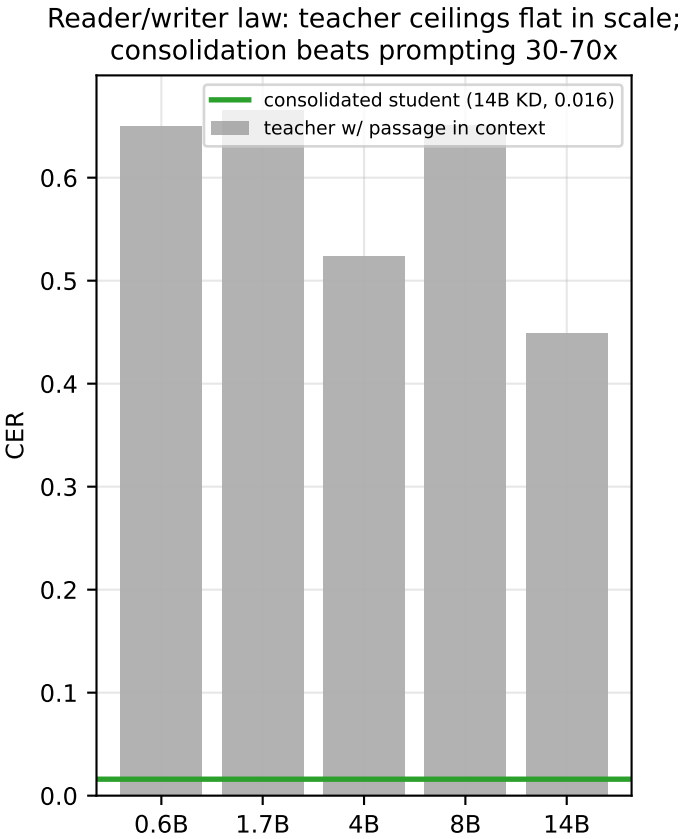


Weight-level identity: the pure-KL arm writes the SAME storage as the best-reciting baseline (peak block and magnitude identical) — only free-run readout sharpening differs



The law: pure distribution-matching converges to teacher fidelity exactly; verbal free-run recall lives in a residual the teacher itself lacks. Teacher-forced fidelity is high (96.5% measured on the PP2 replication) and the stored weights are indistinguishable from the reciting arm's — the residual is a free-running lock-on phenomenon at small scale, NOT missing knowledge. The classic_kd branch provides the complement: at 30B-A3B the same pure objective closes the residual completely (CER 0.000). Scale, not labels, appears to buy the last 3%.

7. layerwise — reader/writer law, dilution, and scale state



Parallelism bench (0.6B, seq 600, hot loop)

single GPU 57.9 ms/item
PP2 (pipeline) 61.4 ms/item (+6%)
TP2 (tensor) 99.7 ms/item (+72%)

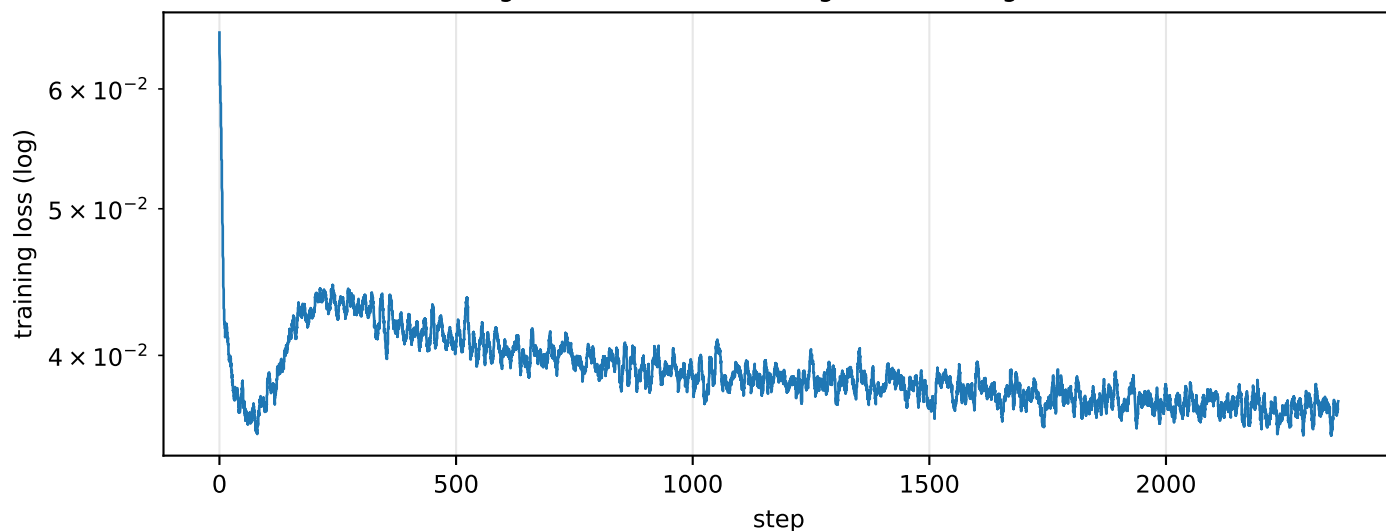
offload_adam: full-FT 4B at 2.3 s/item
hot loop is sync-bound: 1.46x recoverable;
batching+sync fix projected 4-6x (C3 item 5)

PP2 TRAINING repro: pending certification
(the earlier 'failure' was the mid-campaign
default confound, not a parallelism defect)

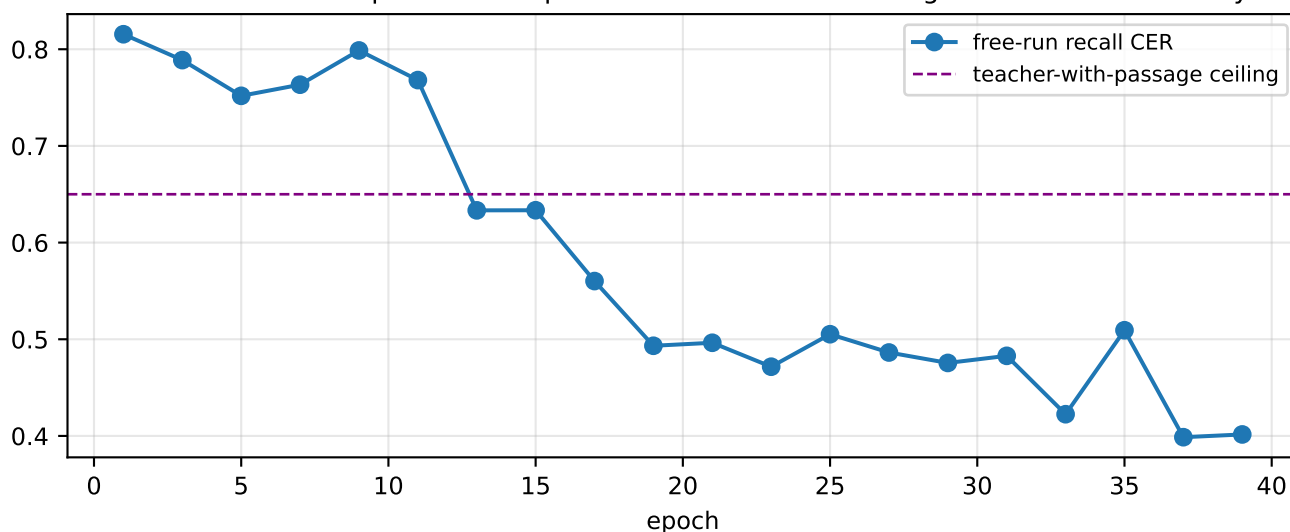
Top-left: at every measured size, the base model READING the passage in context recites far worse (CER 0.45-0.67) than a student that consolidated the text into weights — prompting is not a substitute for consolidation, and the gap does not close with teacher scale. Top-right: co-residence of more memorized content is free or better (dilution). Bottom: the scale state honestly labeled — the sub-0.01 small-scale points are supervised baselines; the pure-recipe-at-scale question is Campaign 3's opener, with classic_kd's 30B result as the existence proof.

8. layerwise — dynamics of the pure teacher-KL arm (lw_r_slide8kl_0p6b_rag)

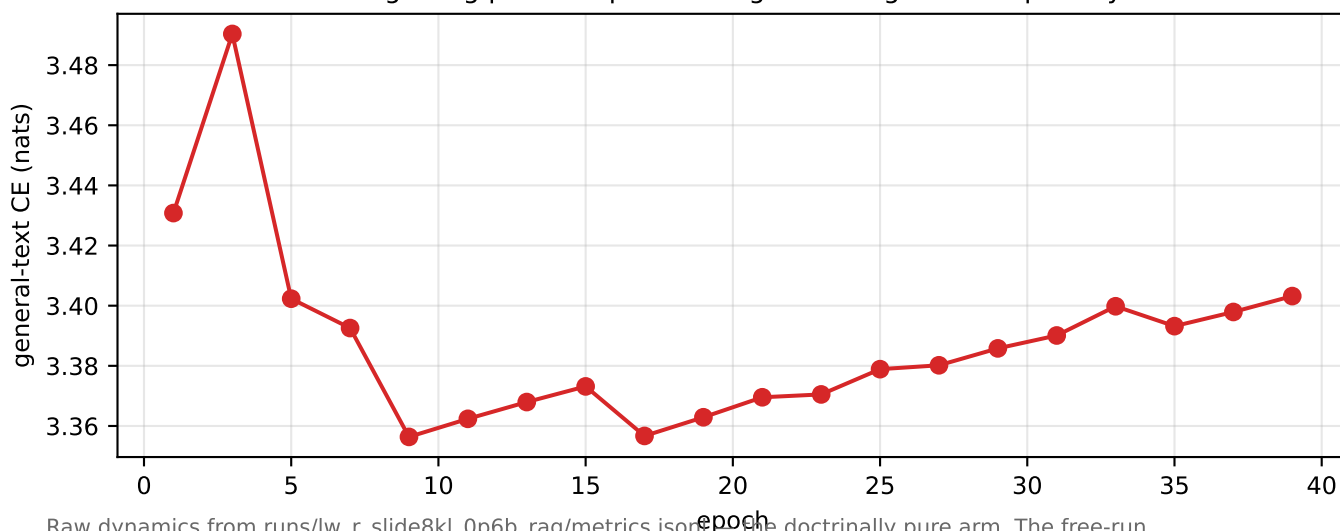
Training loss — smooth convergence; nothing is broken



Free-run recitation probe — the plateau IS the result: convergence to teacher fidelity

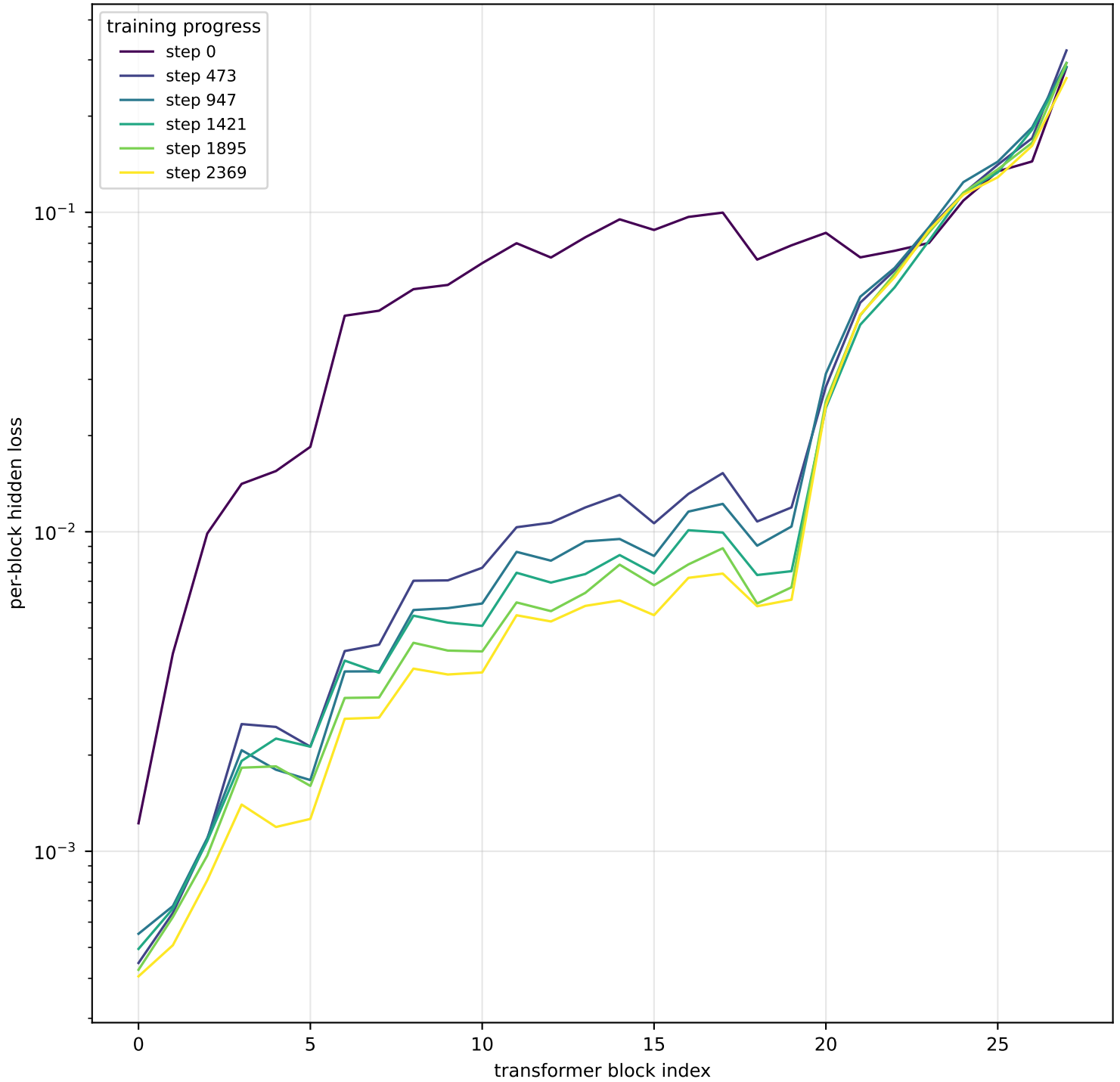


Forgetting probe — pure KL is gentle on general capability



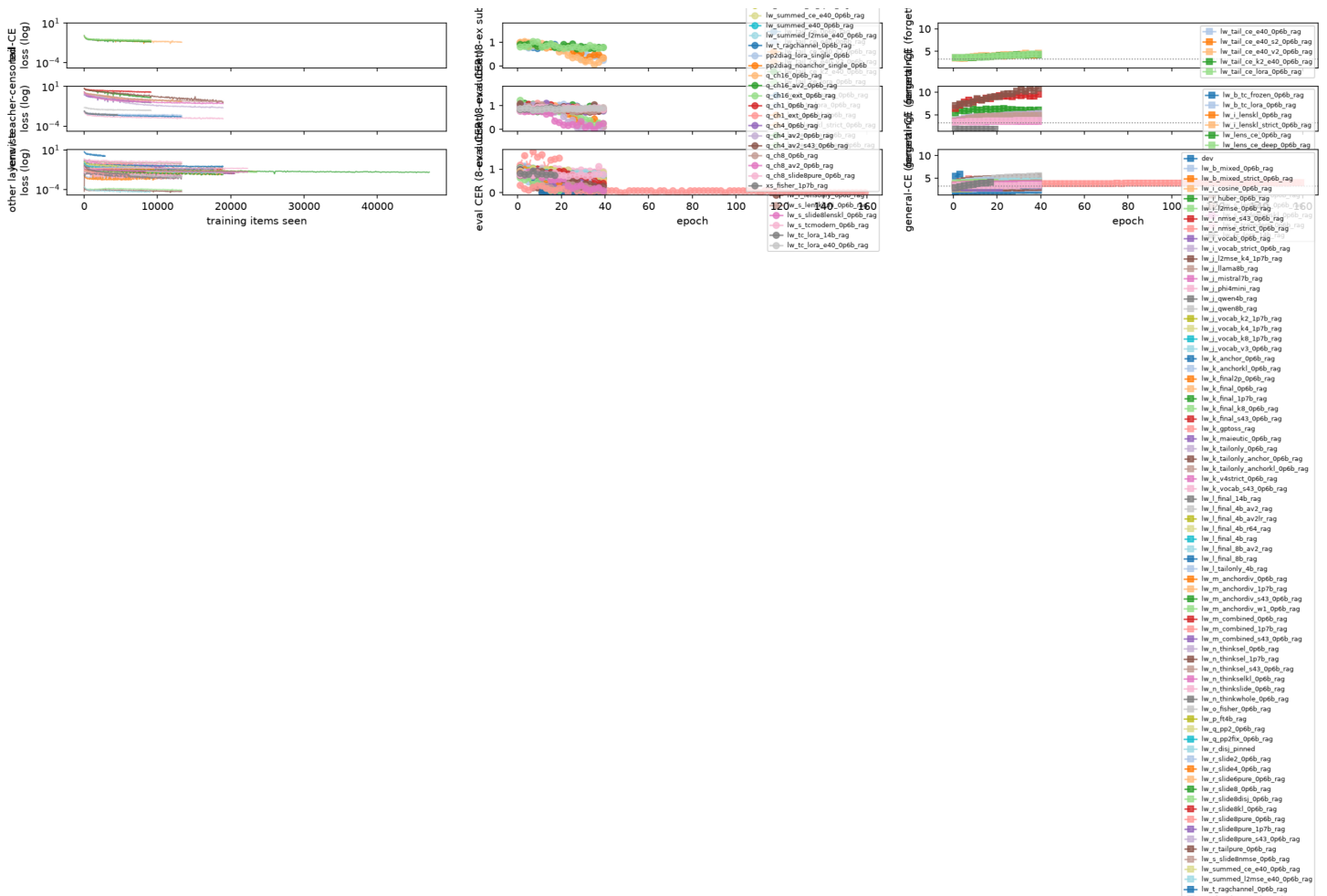
Raw dynamics from runs/lw_r_slide8kl_0p6b_rag/metrics.json — the doctrinally pure arm. The free-run plateau near the teacher ceiling is not a training failure: teacher-forced fidelity is high, the weight-space storage is identical to the reciting baseline's, and the destruction battery is nearly clean. What is missing at 0.6B is only free-running lock-on — the last-3% residual that the 30B classic_kd run shows closing with scale.

9. layerwise — per-block hidden-loss profile over training



Each curve is the per-block hidden-matching loss at one moment of training (dark early, light late). The persistent upward slope toward deep blocks is the divergence signature of attention into the privileged block: deeper layers carry more context-dependent state — exactly the signal being distilled. Training compresses the profile by roughly an order of magnitude while preserving its shape: storage is written everywhere, not only where the loss is largest (single-layer ablations at the delta peak are harmless; the redundancy is causal, not apparent).

10. layerwise — full campaign curve board (generated artifact)



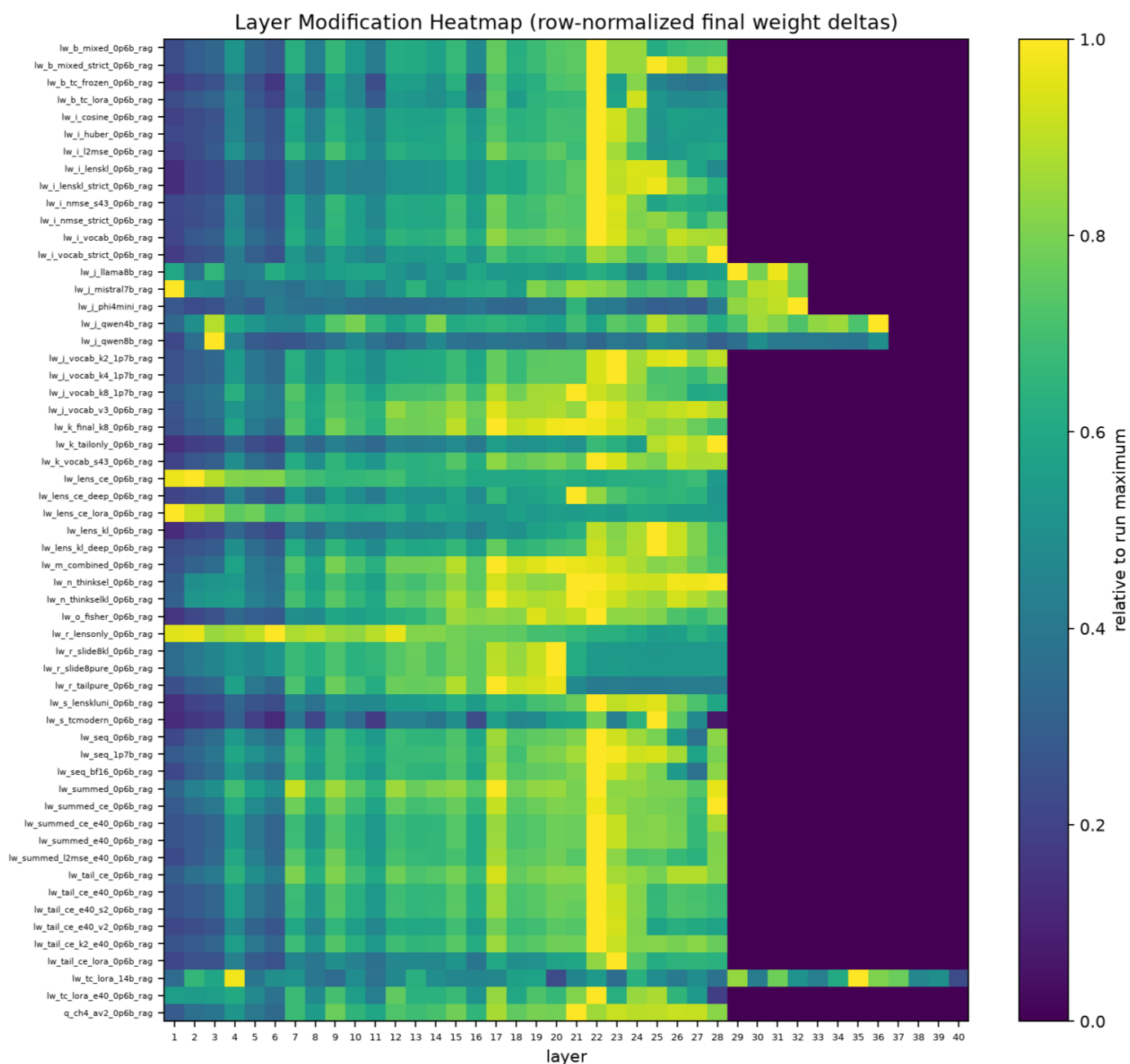
runs/curves.png as maintained by the branch's own report pipeline: recitation CER, line-exact and general-CE trajectories for the active arms of the final waves. Sub-0.01 tracks are supervised-baseline arms; the pure-KL family is the plateau band per the last-3% law.

11. layerwise — forgetting curve board (generated artifact)



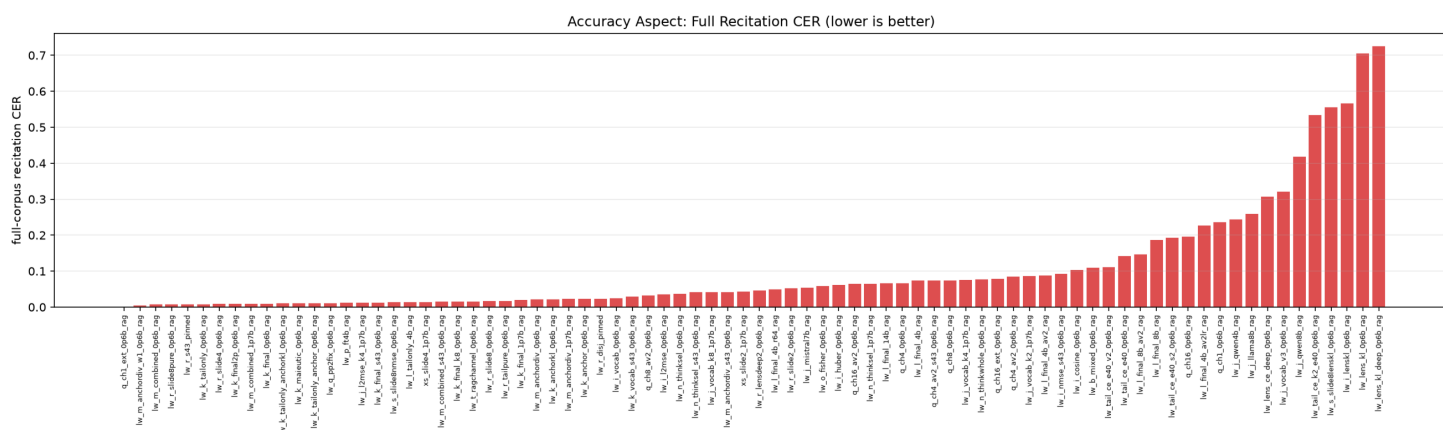
runs/forget_curves.png: per-category general-text cross-entropy drift during training for the destruction-battery categories — the instrument behind the Anchor-Goodhart law (C2-2/3): protection follows the anchor set, so anchor diversity is a free lunch.

12. layerwise — layer modification heatmap (generated artifact)



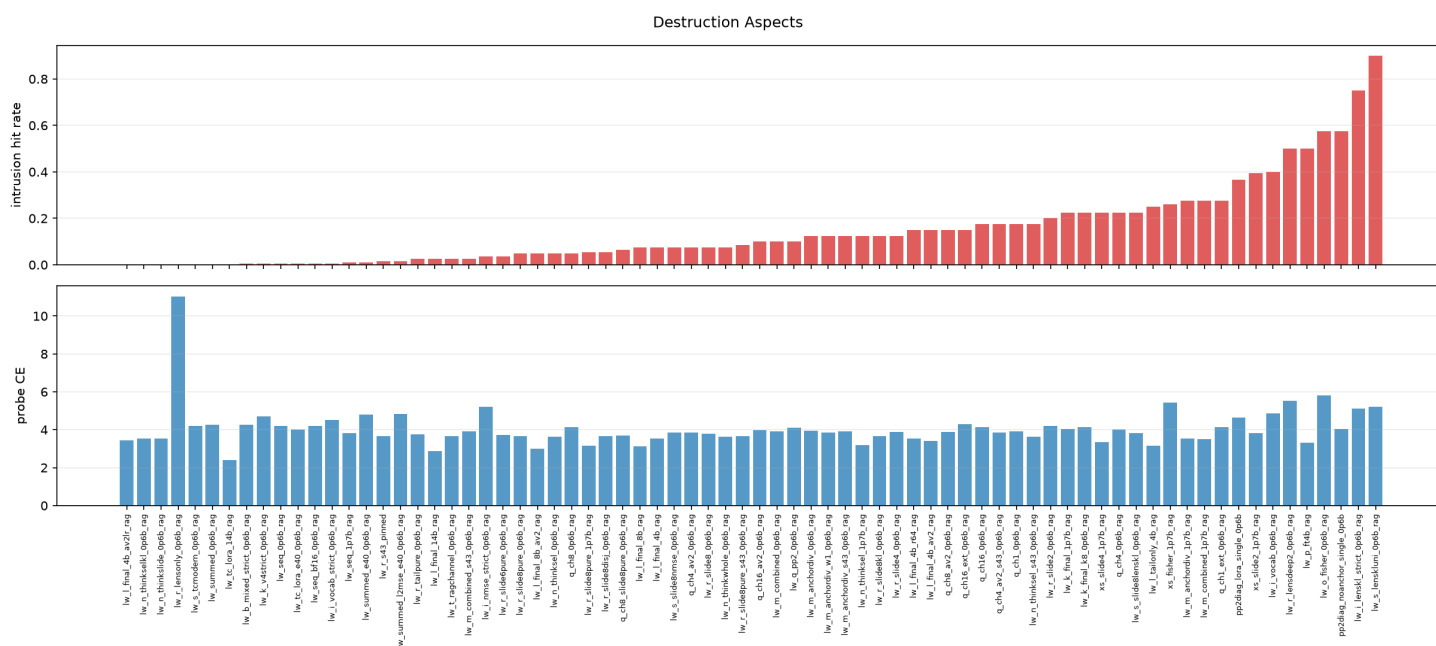
runs/layer_modification_heatmap.png: weight-delta magnitude by block across arms. Delta mass concentrates in the upper-middle stack (~80% depth) in every arm regardless of loss or readout — the storage band that the classic_kd branch independently finds at 14B-30B scale.

13. layerwise — accuracy aspects (generated artifact)



runs/accuracy_aspects.png: recall quality decomposed by aspect (recitation, dialogue elicitation, chain depth, maieutic frames) for the closing-slate arms; gray-classify the supervised-readout arms when reading.

14. layerwise — destruction aspects (generated artifact)



runs/destruction_aspects.png: the category-resolved destruction battery (poetry/prose/procedural/facts/English prose, benchmarks, intrusion) — Campaign 2's verdict machine. Note the pure-KL arms sit near the clean edge of every panel.

15. classic_kd — scope, doctrine and current results

branch classic_kd · 4 principal runs · teacher-logit KL only

SCOPE. This branch studies classical KL-based self-distillation ONLY — the traditional method, implemented honestly, with no label leakage. Standing question: under classical KD, which transformer layers are modified, which layers make the memorized text readable, and how does that localization move with scale?

DOCTRINE. Training is teacher-logit KL only. Any objective that reads corpus tokens as labels is supervised fine-tuning, not distillation, and is out of scope here. Pure top-k KL can saturate while free-run recitation remains poor; that is a RESULT to measure, not a reason to add supervised labels — the same phenomenon the layerwise branch characterizes as the last-3% law, discovered independently on each side of the wall.

CURRENT LESSONS (EXPERIMENTS.md, 2026-07-05):

- The 8-example training-eval subset is front-of-poem biased; only full-corpus eval supports conclusions.
- LoRA needs $lr \sim 1e-4$; $1e-5$ plateaus and produces false negatives.
- Forgetting tracks how much text the model has actually internalized; compare recipes at matched recitation quality, not matched epochs.
- Localization must be read with several probes at once: weight deltas, adapter norms, logit lens, graft/ablate.
- Thinking traces are an ablation, not a default target: compare RAG-hidden-only against RAG+trace-hidden at matched budget before deciding whether reasoning text should be written into weights.

RESULTS SNAPSHOT (runs/current_report.md, generated 2026-07-05 13:02 CEST; base 0.6B control CER 0.932, general NLL 3.278):

Qwen3-30B-A3B-Instruct (MoE, LoRA 3.16% trainable): PERFECT probe recitation reached repeatedly — CER 0.000, line-exact 1.000, best at epoch 11; general-NLL drift +0.22 at epoch 33. Top modified layers L35-L40 of 48. THE PROGRAM'S STRONGEST PURE RESULT: classical teacher-logit KL, no labels anywhere, verbatim recall closed completely.

Qwen3-14B (LoRA 0.43% trainable, 60 epochs done): best CER 0.016 at epoch 27, line-exact 0.969; recall strong but noisier than the MoE run, general-NLL drift the largest of the slate (+0.35). Top modified layers L29-L36 of 40.

Qwen3.6-27B (LoRA 0.30% trainable, training, epoch 6 at snapshot): CER improved sharply to 0.459 by epoch 3, not yet stabilized. Top modified layers L50-L55.

kd_full_0p6b (full-FT pure-KD control): drives KL low but recitation stalls near CER 0.64 at epoch 9 — the small-scale saturation result, exactly matching the layerwise branch's last-3% plateau at the same scale.

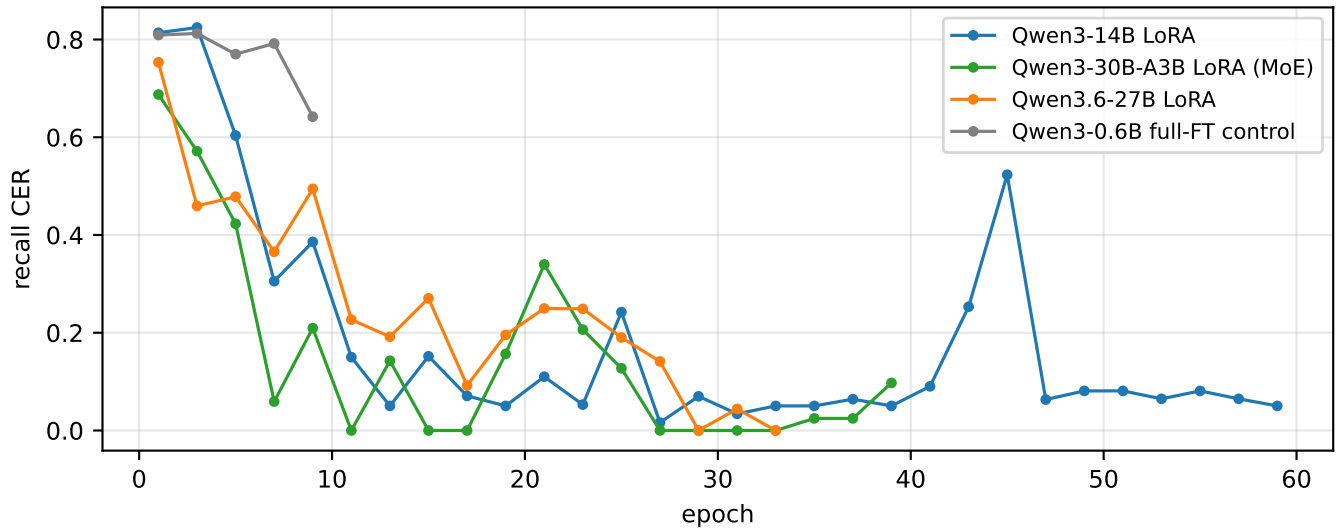
THE SCALE READING. Pure KL closes the verbatim residual as the student grows: 0.6B stalls (~ 0.64 -0.80 across both branches), 14B reaches 0.016, 30B-A3B reaches 0.000. The 'last 3%' appears to be bought by scale (or by MoE parameter count per item — the dilution law's prediction), not by labels. This single column of numbers is the program's best argument that no supervised concession is needed at the Don Quijote stage.

LOCALIZATION READING SO FAR. In each LoRA run the top modified layers sit at roughly 70-85% of depth (L29-36/40 at 14B, L50-55/ ~ 64 at 27B, L35-40/48 at 30B-A3B) — consistent with proportional rather than absolute depth, and strikingly consistent with the layerwise branch's causally-probed storage band ($\sim 80\%$ depth at 0.6B). Two independent methods, one localization story.

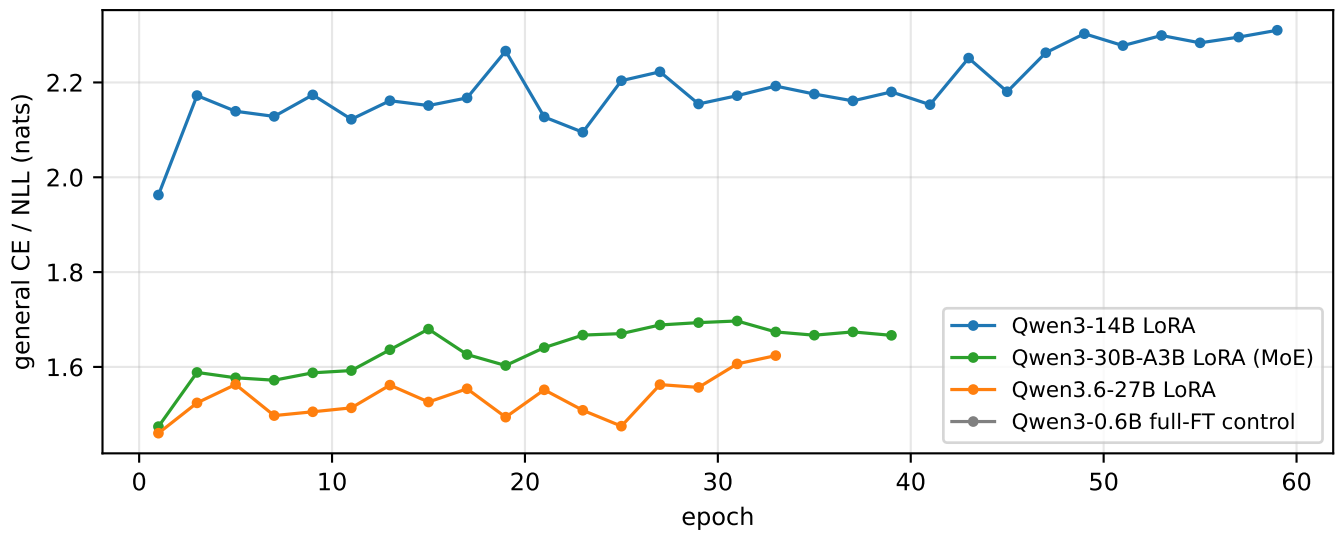
MODEL LADDER (planned): Llama-3.2-1B and SmolLM3-3B for tokenizer/data sanity, Qwen3-4B/8B for the absolute-vs-proportional depth question, DeepSeek-V2-Lite as first expert-routing MoE, R1-Distill-Qwen-1.5B for the thinking-hiding arm, then the 4xL40S ladder toward the H100 Don Quijote stage.

16. classic_kd — training dynamics of the KD slate

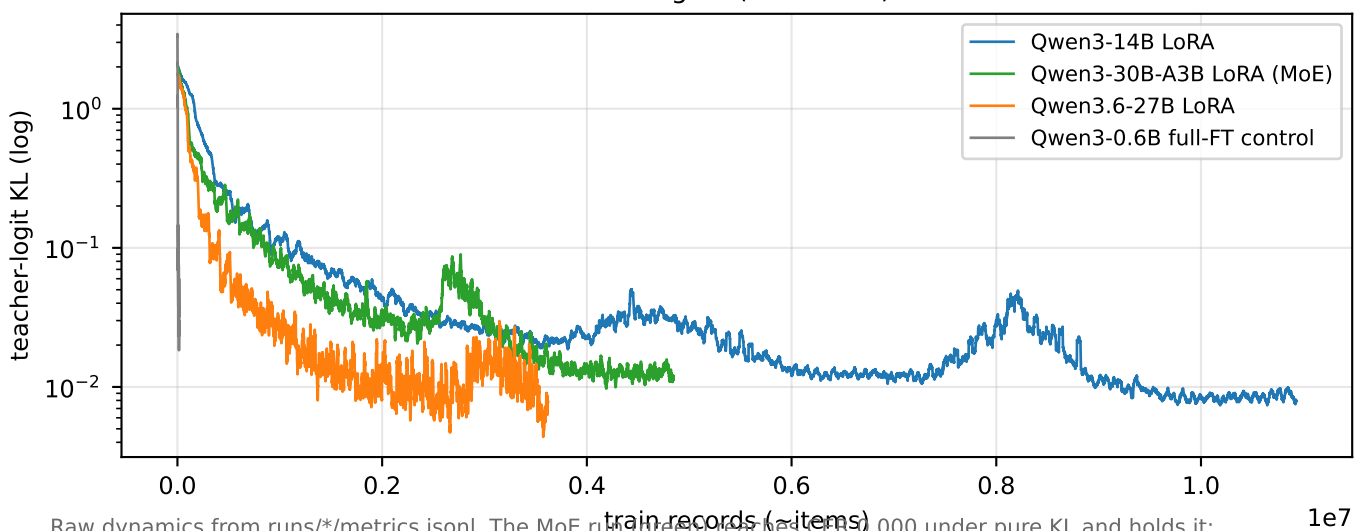
Free-run recitation probe (full poem)



Forgetting probe (held-out general text)



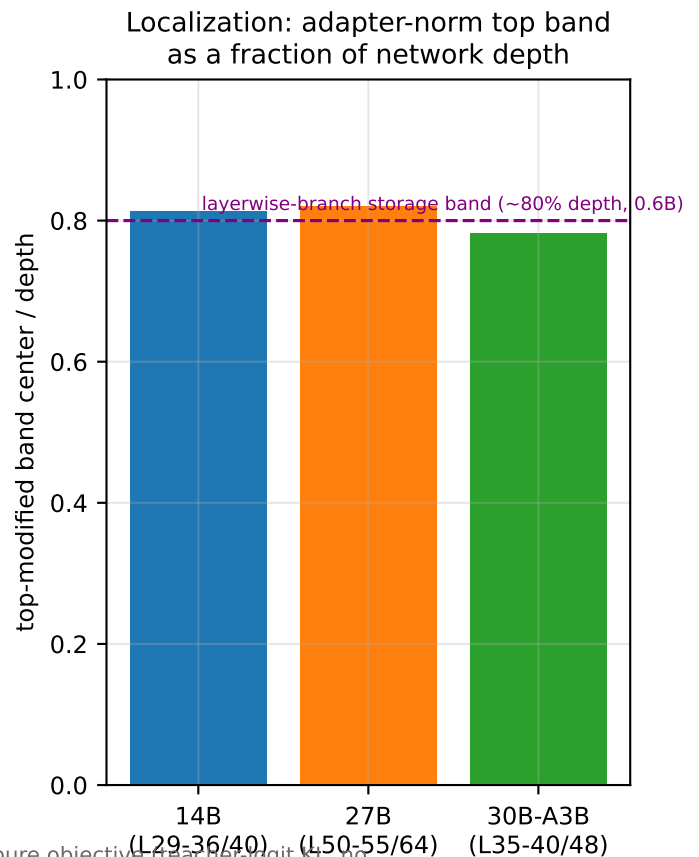
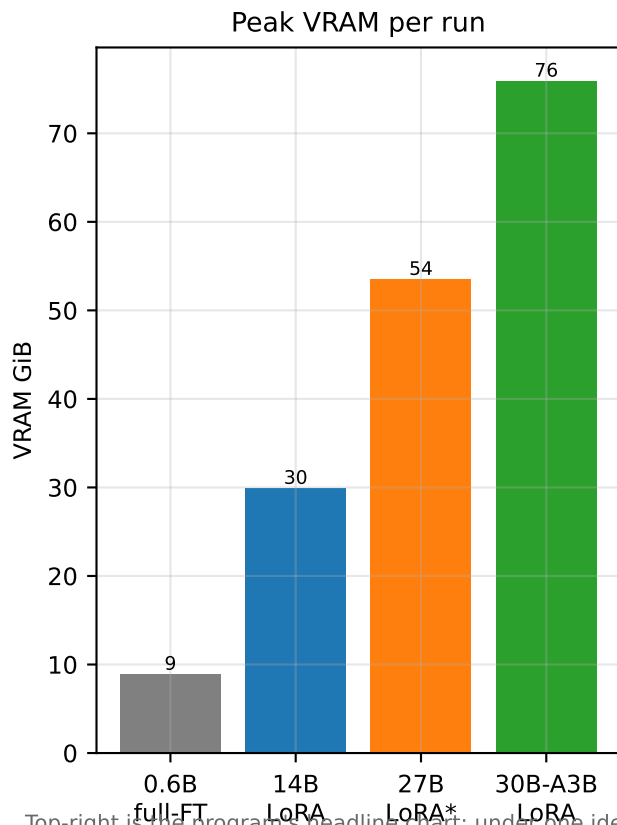
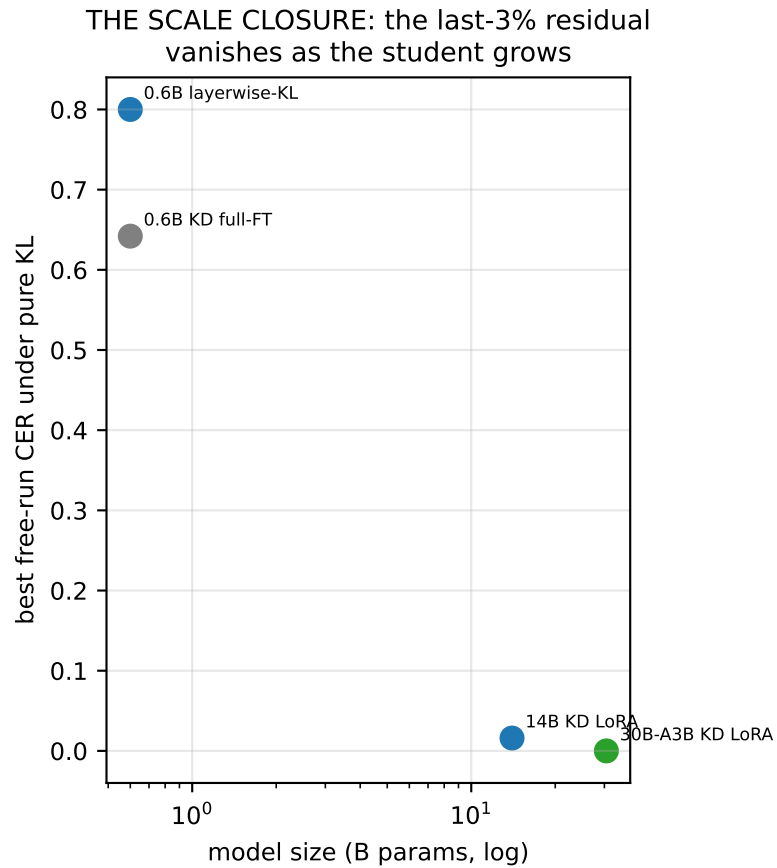
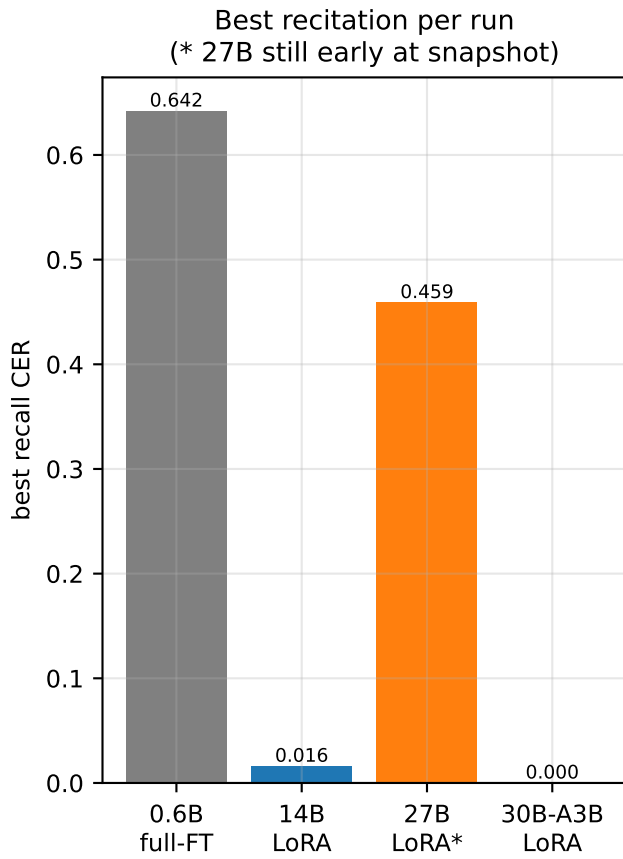
Training KL (smoothed)



Raw dynamics from runs/*/metrics.jsonl. The MoE run (green) reaches CER 0.000 under pure KL and holds it; 14B (blue) converges more noisily to 0.016; 27B (orange) was 6 epochs in at snapshot. The gray full-FT 0.6B control is the small-scale saturation result: KL falls smoothly, recitation stalls — the same plateau the layerwise branch measures at this scale. Gen-NLL baselines differ per model; drift within a run, not absolute level across runs, is the forgetting signal.

1e7

17. classic_kd — slate summary, scale closure, localization



Top-right is the program's headline chart: under one identical pure objective (teacher-logit KL, no labels), free-run verbatim recall goes 0.80 -> 0.64 -> 0.016 -> 0.000 as the student scales from 0.6B to 30B. Bottom-right: the most-modified band sits at 70-85% of depth at all three scales, favoring PROPORTIONAL over absolute depth and agreeing with the layerwise branch's causally-probed storage band. The 4B/8B ladder runs will sharpen both claims.

18. Synthesis — two branches, one picture

compiled 2026-07-06 · sources: EXPERIMENTS.md (both branches), runs/results.md, runs/current_report.md,

CONVERGENT FINDINGS (independently derived on both sides of the doctrinal wall):

1. WHERE IT WRITES. Layerwise (causal probes: ablation, grafting, delta profiles) locates storage in a redundant upper-middle band (~80% depth). Classic KD (adapter-norm localization at 14B/27B/30B) finds its most-modified band at 70-85% of depth across scales. Two methods, two branches, one localization story — and it looks proportional, not absolute.
2. STORING IS NOT SPEAKING — AT SMALL SCALE. Layerwise: strict hidden matching stores signal but never recites; pure teacher-KL arms converge to teacher fidelity exactly with storage bitwise-comparable to reciting arms (the last-3% law, seven replications). Classic KD: pure KL at 0.6B saturates while recitation stalls at 0.64. Same phenomenon, discovered independently, stated in each branch's own vocabulary.
3. THE RESIDUAL CLOSES WITH SCALE. Under the identical pure objective, best free-run CER goes 0.80 (0.6B layerwise-KL) -> 0.64 (0.6B KD) -> 0.016 (14B KD) -> 0.000 (30B-A3B KD). No labels were needed anywhere in that column. This is the program's central encouraging fact: the supervised concessions explored in the layerwise campaigns (tail-CE / task-CE readouts — per the owner's classification, traditional KL camouflaged as innovation) appear to be small-scale crutches, not necessities. What they bought at 0.6B, scale buys honestly at 14B+.
4. CONSOLIDATION BEATS PROMPTING. Teacher ceilings with the passage in context are flat at 0.45-0.67 CER from 0.6B to 14B, while consolidated students reach 0.016-0.000. Writing to weights outperforms reading from context by 30-70x at every measured size, and the gap does not close with teacher scale.

WHAT EACH BRANCH STILL OWES:

classic_kd: a destruction battery. The 30B-A3B champion's CER 0.000 has no category-resolved damage profile and no intrusion-bait measurement yet; Campaign 2 on the layerwise side showed exactly such headline numbers can hide substantial intrusion. Cleanliness is UNMEASURED, not established. Also owed: the 4B/8B localization pair (absolute-vs-proportional becomes a claim there), DeepSeek-V2-Lite expert localization, and the thinking-censorship ablation at matched budget.

layerwise: the pure recipe at scale. Bounded backward depth (block-local credit) is what makes 120B-class streamed-block training thinkable — classic KD cannot offer that; it backpropagates through everything. Campaign 3 must show hidden-matching + teacher-KL readout at 1.7B-27B rides the same scale closure the KD branch measured, at a fraction of the backward cost. Also owed: the two pinned confound-resolution twins (seed claim; disjoint-window claim), teacher-stream k-windows, premise-gated thinking, and the 4-6x hot-loop fix.

Shared summit: a 120B-class model that has actually memorized Don Quixote and can still do everything else — with the classic branch defining what honest distillation achieves, and the layerwise branch making it affordable blockwise.

REPRODUCIBILITY NOTE. Both branches carry the same hard-won meta-lesson, now enforced in tooling on the layerwise side: config DEFAULTS are experiment variables. Two silent-default confounds in one day each produced valid science only because every run dumps its full config — the dump is the lab notebook; the unset-sentinel validator makes it sufficient.